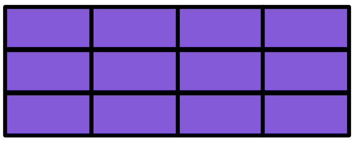
Formalising a sequence

Task A: Pictorial sequences

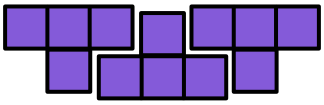
1) Andeep is making different patterns that go up in fours.

Alex is making different patterns that go up in threes.

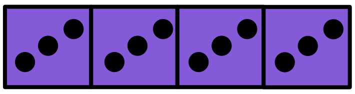
They show one image from each of their patterns. Which are Andeep's, which are Alex's and which could be either?

a) 

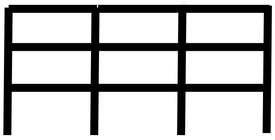
Either

b) 

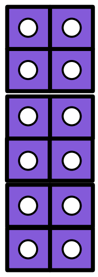
Andeep

c) 

Alex

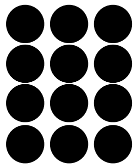
d) 

Either depending on whether we refer to total lines or just horizontal lines

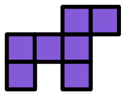
e) 

Andeep

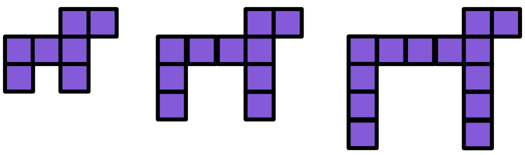
Either

f) 

Sam has drawn a 'dog' by shading in squares. Their classmates each continue the pattern in different ways.



2) Aisha continues it this way

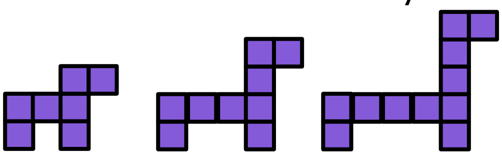


a) Describe how the pattern is developing in words

E.g. The body is getting longer by 1 and each leg is getting longer by 1.

b) How many squares would be needed for the 10th dog? *34*

3) Lucas continues it this way

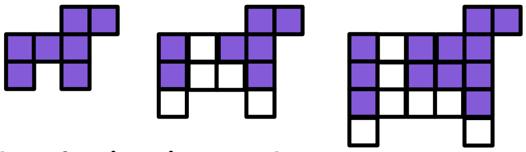


a) Describe how the pattern is developing in words

E.g. The body is getting longer by 1 and the neck is getting longer by 1

b) How many squares would be needed for the 10th dog? *25*

4) Izzy continues it this way



a) Describe how the pattern is developing in words.

E.g. The legs are both getting longer by 1 each time (2 in total). The body increases in both directions. A 1 by 1 square, then a 2 by 2 square then a 3 by 3 square

b) How many squares would be needed for the 10th dog?

The 10th dog will have a 10 by 10 square in the middle 11 squares for each leg and 2 for the head. 124 squares

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Task B: Describing patterns in pictorial sequences

1) For each sequence, fill in the table for the number of squares and perimeter. Describe the rule for both sequences.

a)

Term	1	2	3	4	5
Squares	1	2	*3*	*4*	*5*
Perimeter	4	6	*8*	*10*	*12*

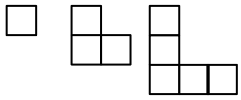
Rule for squares:

Increase by 1 each time/ Same as the term number

Rule for Perimeter:

Increase by 2 each time/ 2 lines for each square plus 2 on each end

b)



Term	1	2	3	4	5
Squares	1	*3*	*5*	*7*	*9*
Perimeter	4	*8*	*12*	*16*	*20*

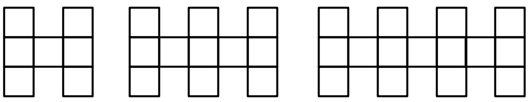
Rule for squares:

Increase by 2 each time/ 2 squares per diagonal minus 1 for first diagonal

Rule for Perimeter:

Increase by 4 each time/ Term number multiplied by 4

c)



Term	1	2	3	4	5
Squares	7	*11*	*15*	*19*	*23*
Perimeter	*16*	*24*	*32*	*40*	*48*

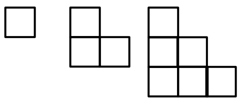
Rule for squares:

Increase by 4 each time/ 4 squares per H plus 3 for the start

Rule for Perimeter:

Increase by 8 each time

d)



Term	1	2	3	4	5
Squares	*1*	*3*	*6*	*10*	*15*
Perimeter	*4*	*8*	*12*	*16*	*20*

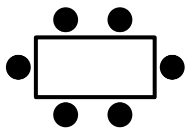
Rule for squares:

Increase by 2 , then 3 then 4.../ Add term number on to the previous term

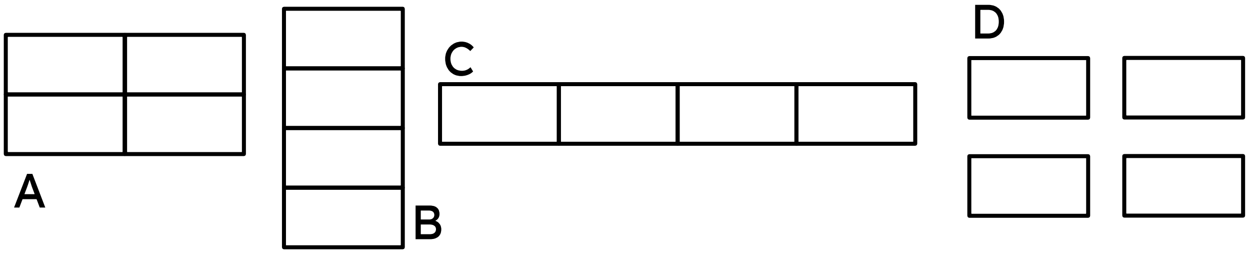
Rule for Perimeter:

Increase by 4 each time/ Term number multiplied by 4

2) In a school dining hall, 6 chairs (circles) fit around one rectangular table



Below are 4 ways to arrange 4 tables



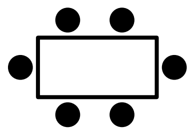
a) To calculate the number of chairs needed, the caretaker multiplies the number of tables by 6. Which way are they planning on arranging the tables?

Arrangement D

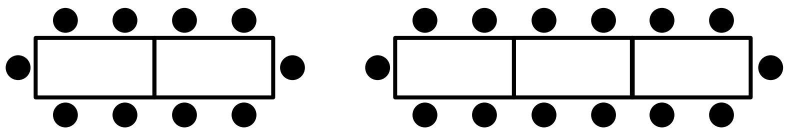
b) For each arrangement how many more chairs can be added if an extra table is added?

A: 2 or 4 depending which way round, B: 2, C: 4, D:6

3) In a school dining hall, 6 chairs (circles) fit around one rectangular table



The tables are arranged like this



a) How many chairs will fit round 4 tables? *18*

b) What is the relationship between the number of chairs and the number of tables?

Each table can have 4 chairs plus 2 for the ends

c) How many chairs can fit round 10 tables? *42*

d) Jacob has 25 classmates. How many tables are needed so all 26 of them can have a chair. *6 tables*